

Setup QNX dependencies

To run **QNX build** from **command line** it is mandatory to have installed:

- **QNX Software center**
- **SDP 7.1**
- **Qualcomm patches**
- **Libraries needed by Qualcomm BSP**
- **Libraries needed to develop Marelli custom software**

Momentix IDE is needed to debug or to write new QNX Application using templates, but it is not mandatory to just compile.

Configuring QNX build environment

Follow these steps to setup QNX build environment

1. Install **QNX Software center**
2. Install **SDP** and **Qualcomm BSP**
3. Install, if needed, **Momentix IDE**

Installing QNX SW Center

Using command line

1. register to **QNX portal** on <https://www.qnx.com/account/login.html>
 - a. use [Marelli address](#)
2. contact SPL or IT and ask to add your account to DCU project
 - a. so you'll be able to use one (of the few) **Floating licenses** available for the project
3. Download **QNX SW Center** for *Linux* host from <https://www.qnx.com/download/feature.html?programid=58067>
4. Follow installation notes - https://www.qnx.com/developers/articles/inst_6861_1.html
5. run the file and follow on screen prompt to install it
6. cd to path where **QNX SW Center** was installed, you'll find executable **qnxsoftwarecenter_clt**
7. export QNX username and Password

```
a. export MYQNX_USER=name.surname@marelli.com
   export MYQNX_PASSWORD=myPassword
```

8. Activate licenses

```
a. ./qnxsoftwarecenter_clt -syncLicenseKeys -myqnx.user="$MYQNX_USER" -myqnx.password="$MYQNX_PASSWORD" -activateAll -listLicenseKeys
```

9. Mirror repositories

```
a. ./qnxsoftwarecenter_clt -mirror -myqnx.user="$MYQNX_USER" -myqnx.password="$MYQNX_PASSWORD"
```

Using GUI

1. register to **QNX portal** on <https://www.qnx.com/account/login.html>
 - a. use Marelli address
2. contact SPL or IT and ask to add your account to DCU project
 - a. so you'll be able to use one (of the few) **Floating licenses** available for the project
3. Get **QNX Software center** from QNX portal and install it



For more details refer to QNX official documetation:

- QNX_Software_Center_User_Guide_202105270358.pdf
- QNX_Floating_License_Management_20210218.pdf

Documents can be found in <https://www.qnx.com/developers/docs/index.html>

Installing SDP and dependencies

Two ways are available to get **SDP** (and needed dependencies):

- use an installation bundle **suggested method**
- manual installation (commandline or gui)

Using bundle

SDP bundle archive is provided by Marelli System Integration team and **will contain all needed dependencies** (Qualcomm dependencies + Marelli dependencies)

SDP bundle is for Linux build environment

1. Download all the archives from Artifactory https://artifactory.marelli.com/ui/repos/tree/General/CCSP/CCSP_tools/DCU/qnx/sdp/sdp_7.1-qc_0000201.1a-mm_0.1
2. please refer to table **Version Reference** below to get correct bundle version depending on Qualcomm release
3. update Artifactory links as per **Version reference** to get in to your Linux machine using curl, **below is an example for qc_0000201.1a**
4. **Downloading and unzipping the artifacts can be done in any path.**

```
curl -u <Username>:<password> -O "https://artifactory.marelli.com/artifactory/CCSP/CCSP_tools/DCU/qnx/sdp/sdp_7.1-qc_0000201.1a-mm_0.1/sdp_7.1-qc_0000201.1a-mm_0.1.001"
curl -u <Username>:<password> -O "https://artifactory.marelli.com/artifactory/CCSP/CCSP_tools/DCU/qnx/sdp/sdp_7.1-qc_0000201.1a-mm_0.1/sdp_7.1-qc_0000201.1a-mm_0.1.002"
curl -u <Username>:<password> -O "https://artifactory.marelli.com/artifactory/CCSP/CCSP_tools/DCU/qnx/sdp/sdp_7.1-qc_0000201.1a-mm_0.1/sdp_7.1-qc_0000201.1a-mm_0.1.003"

ex: curl -u <FXYZABC>:<urpassword> -O "https://artifactory.marelli.com/artifactory/CCSP/CCSP_tools/DCU/qnx/sdp/sdp_7.1-qc_0000201.1a-mm_0.1/sdp_7.1-qc_0000201.1a-mm_0.1.001"
```

1. use **7z** to *reconstruct* single archive
 - a. e.g.

```
7z x sdp_7.1-qc_0000201.1a-mm_0.1.001
```

2. unzip the .zip archive **sdp_7.1-qc_0000201.1a-mm_0.1.zip**
 - a. e.g.

```
unzip sdp_7.1-qc_0000201.1a-mm_0.1.zip
```

3. the folder you get is **SDP installation folder**
 - a. This path can later be used as reference in the **command line build**
 - i. please refer to section [BuildingQNX](#), using flag **--external PATH_TO_JUST_EXTRACTED_FOLDER**
 - b. or with **Momentix**
 - i. go to Preferences QNX
 - ii. on SDK Selection press "Add"
 - iii. configure the new SDP installation with the following parameters (just adapt the path to your working area)

SDK Platform Configuration

Import SDK Platform

Specify the path to an SDK platform and a name to uniquely identify this SDK.

Target Path: C:\Users\F52031A\dcu\qnx710\target Browse...

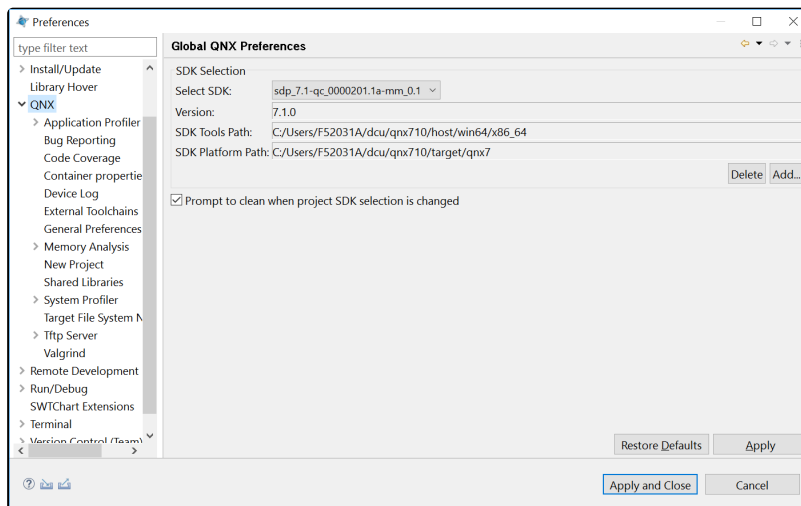
Host Path: C:\Users\F52031A\dcu\qnx710\host Browse...

Name: sdp_7.1-qc_0000201.1a-mm_0.1

Version: 7.1.0

Finish Cancel

- iv.
- v. then just be sure that the new SDK is the default one



vi.



Is it possible to modify bundle?

Yes, it should be, but not suggested... ask **System Integration** to **generate** a **new bundle** with the new mandatory dependency (e.g. new library), so every developer will get the new dependency 😊

~~If you really need to modify the SDP bundle, just follow next section "**Install Marelli dependencies (or additional packages)**" <---not tested.~~

Using manual installation - commandline (Ignore if SDP bundle method is followed)

1. Install SDP 7.1

```
./qnxsoftwarecenter_clt -destination ~/qnx710 -installBaseline com.qnx.qnx710 -myqnx.user="$MYQNX_USER" -myqnx.password="$MYQNX_PASSWORD" -flexLicenseServer 27009@ittor47lic3.mmemea.marelliad.net
```

2. clone repo containing Qualcomm proprietary code (Marelli mirror)

```
git clone https://hub.marelli.com/bitbucket/scm/ccsp_soc/sa8295p-hqx-4-2-4-0_hlos_dev_qnx.git --depth=1
```

3. cd to the path where you've clone sa8295p-hqx-4-2-4-0_hlos_dev_qnx
4. Install Patchset from Qualcomm
 - a. in this way ~/qnx710 will be updated with mandatory packages

```
~/qnx/qnxsoftwarecenter/qnxsoftwarecenter_clt -destination ~/qnx710 -myqnx.user="$MYQNX_USER" -myqnx.password="$MYQNX_PASSWORD" -flexLicenseServer 27009@ittor47lic3.mmemea.marelliad.net -importAndInstall apps/qnx_ap/qnx_bins/prebuilt_HYP710FIXME/patchset/EXP_Gen3_Gen4_SDP71_HV22_20220301.plist
```



Different version of SDP will require different intall command (e.g. ./qnxsoftwarecenter_clt -destination ~/qnx700 -installBaseline com.qnx.sdp/7.0 -myqnx.user="\$MYQNX_USER" -myqnx.password="\$MYQNX_PASSWORD")

- b. verify installation:
 - i. use md5 command; (note: it'll shown message only in case of error)

```
md5sum --quiet -c /opt/builds/alessio/dcu/qnx_test/sa8295p-hqx-4-2-4-0_hlos_dev_qnx/apps/qnx_ap/qnx_bins/prebuilt_HYP710FIXME/patchset/hyp710_hash.md5
```

5. Install Marelli dependencies (or additional packages)
 - i. Search needed packages

```
~/qnx/qnxsoftwarecenter/qnxsoftwarecenter_clt -list -myqnx.user="$MYQNX_USER" -myqnx.password="$MYQNX_PASSWORD" -flexLicenseServer 27009@ittor47lic3.mmemea.marelliad.net
```

- ii. Install it (opencv example):

```
~/qnx/qnxsoftwarecenter/qnxsoftwarecenter_clt -destination ~/qnx710 -myqnx.user="$MYQNX_USER" -myqnx.password="$MYQNX_PASSWORD" -flexLicenseServer 27009@ittor47lic3.mmemea.marelliad.net -installPackage com.qnx.qnx710.osr.opencv
```

Using manual installation - gui (Ignore if SDP bundle method is followed)

To install QNX SDP 7.1 (development kit) and Momentix (IDE to develop for QNX) one need to have proper license:

- a. **SDP 7.1**
- b. if you get license error it means that Project manager haven't granted you usage of the floating license
- c. Install Patchset from Qualcomm
 - i. clone repo containing Qualcomm proprietary code (Marelli mirror)
 - ii. ~~git clone https://hub.marelli.com/bitbucket/scm/ccsp_qc_re/sa8295p-hqx-4-2-4-0_hlos_dev_qnx.git --depth=1~~
 - iii.

```
git clone https://hub.marelli.com/bitbucket/scm/ccsp_qc_re/sa8295p-hqx-4-2-4-0_hlos_dev_qnx.git --depth=1
```
 - iv. From QNX SW center **import** the **patchset** from file apps/qnx_ap/qnx_bins/prebuilt_HYP710FIXME/patchset/EXP_Gen3_Gen4_SDP71_HV22_20220301.plist
 - v. From QNX SW center **manually install all needed libraries**
2. Before running Momentix you have to set the correct license server (that is reachable from Marelli network)
 - a. go to **SDP 7.1** install folder (e.g. C:\Users\FXXXX\qnx710)
 - b. open command prompt
 - c. run **qnxsdp-env.bat** (or the sh script if running on Linux)
 - d. run command

```
lmutil lmpath -override qnxlm 27009@ittor47lic3.mmemea.marelliad.net
```

Version Reference

Qualcomm release	Release Name	dev_qnx repo commit id	QNX SDP Bundle	Manifest details {Applicable only for android}	dev_la commit id
r0000201.1a (Without Marelli changes)	ES2	a98fc857fe67ea1bcc59748bf8f ae6be6cc46f3f- r0000201.1a	https://artifactory.marelli.com/artifactory/CCSP/CCSP_tools/DCU/qnx/sdp/sdp_7.1-qc_0000201.1a-mm_0.1/sdp_7.1-qc_0000201.1a-mm_0.1.001	-m LA.AU.0.3.1.r3-01400-sa8155_gvmq.0.xml -b refs/tags/mm_r0000201.1a	5791121102b56c80ae01df883e 8d8f5cfe223eeb- r0000201.1a
r00004.2 (Without Marelli changes)	FC1	a150bfc375eeb941d826e685b 33f20a40ffcf07- r00004.2	https://artifactory.marelli.com/artifactory/CCSP/CCSP_tools/DCU/qnx/sdp/sdp_7.1-qc_00004.2-mm_0.1/sdp_7.1-qc_00004.2-mm_0.1.7z.001	-m LA.AU.0.3.1.r3-02200-sa8155_gvmq.0.xml -b refs/tags/LA.AU.0.3.1.r3-02200-sa8155_gvmq.0	369ffe53c0d46012e29717d8bc8f c7caa044c6f8- r00004.2